STUDENT ASSIGNMENT SUBMISSION - PHYSICS LAB REPORT

NexSchool Enterprise Education Platform | Academic Session 2025-2026

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1. Student Identification & Submission Details

Student Name: Mohammad Khan | Admission Number: ADM-2026-0003 | Roll Number: 3
Class: Grade 10 - Section A | Submitted To: Sarah Jenkins (Faculty of Science)
Assignment: Newton Laws & Dynamics Problem Set | Date of Submission: October 12, 2026

2. Solution to Question 1: Net Force & Acceleration

Given: Mass m = 5.0 kg, Applied Force F_app = 35.0 N, Frictional Resistance f_k = 14.0 N.
Formula: F_net = F_app - f_k = 35.0 N - 14.0 N = 21.0 N.
Acceleration: a = F_net / m = 21.0 N / 5.0 kg = 4.20 m/s^2 along horizontal direction.
Status: Verified with dimensional analysis [M][L][T]^-2 / [M] = [L][T]^-2.

3. Solution to Question 2: Incline Force Resolution

Given: 10.0 kg crate on 30 degree frictionless slope. Gravitational acceleration g = 9.80 m/s^2.
Parallel component: F_parallel = m * g * sin(30 deg) = 10.0 * 9.80 * 0.50 = 49.0 N.
Perpendicular normal force: F_N = m * g * cos(30 deg) = 10.0 * 9.80 * 0.866 = 84.87 N.
Conclusion: Acceleration down the incline a = g * sin(30 deg) = 4.90 m/s^2 independent of mass.

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